

Customer Segmentation using RFM Analysis

Project 1 | Data Analysis Internship, Syntecxhub

Name: Rani Rai **Tool:** Microsoft Power BI Desktop **Dataset:** AdventureWorksDW

1. Objective

The goal of this project is to group customers by how recently, how often and how much they buy (RFM analysis), understand how each group behaves, and suggest a targeted marketing action for every group. The work covers the six tasks in the project brief: cleaning the data, calculating RFM metrics, segmenting customers, analysing behaviour, giving marketing recommendations and visualising the result in a dashboard.

2. Executive Summary

- 18,484 customers and about \$29.36 million in sales (July 2005 to July 2008) were analysed and placed into 11 segments.
- Five segments (Promising, At Risk, Need Attention, Loyal and Cannot Lose Them) produce about 95.7% of all revenue.
- At Risk customers are only 9.8% of the base but bring 26.9% of revenue, so they are the most important group to win back.
- New Customers are 18.5% of the base but only 0.44% of revenue. They need a push to make a second purchase.
- 62.9% of customers (11,619) bought only once, so Champions and Loyal are small groups. Growing repeat purchases is the main opportunity.

3. Dataset and Tools

The AdventureWorksDW workbook is Microsoft's public sample data warehouse. Three of its tables were used:

Table	Rows	Used for
FactInternetSales	60,398	Order lines: CustomerKey, SalesOrderNumber, OrderDate, SalesAmount
DimCustomer	18,484	Customer details: gender, yearly income, occupation, education
DimGeography	655	Country and city of each customer

A separate lookup table (*RFM_Table.xlsx*) maps each 3-digit RFM score to a segment name and describes a marketing action for each segment. Tools used: Power BI Desktop with Power Query for cleaning and DAX for the calculations and measures.

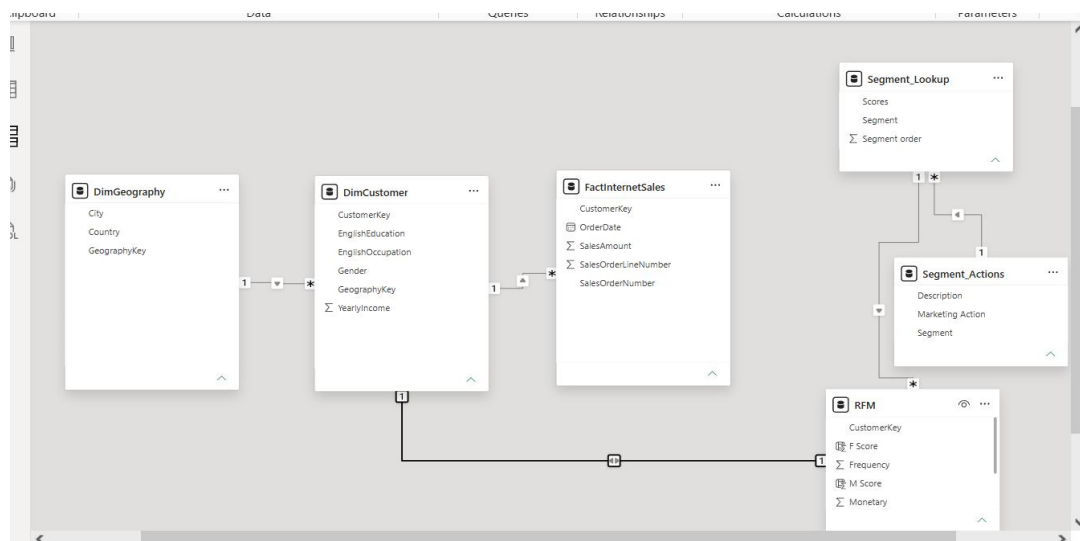
4. Data Cleaning and Preparation

Column profiling was run on the entire data set (not just the top 1,000 rows) in Power Query.

- All columns in FactInternetSales are 100% valid: 0% errors and 0% empty values.
- There are no duplicate order lines (same order number and line number) and no zero or negative sales amounts.
- Only the columns needed for the analysis were kept in each table. OrderDate was set to the Date type and SalesAmount to a decimal type.
- EnglishCountryRegionName in DimGeography was renamed to Country.
- The tables were connected in the data model: DimGeography to DimCustomer, DimCustomer to FactInternetSales, and later DimCustomer to the RFM table.

Issues found in the segment lookup table

- Segment names differed in case between the two sheets (for example "Lost customers" and "Lost Customers"), which would break the join. They were standardised.
- Scores 231, 241 and 251 appeared under two segments (About To Sleep and Hibernating Customers). Each score must map to one segment, so they were kept under About To Sleep.



5. RFM Metrics

Metric	Definition	Calculation in Power BI
Recency	Days since the customer's last order	Reference date minus the customer's latest OrderDate
Frequency	Number of distinct orders	Distinct count of SalesOrderNumber
Monetary	Total amount spent	Sum of SalesAmount

Reference date: 1 August 2008, which is the last order date in the data (31 July 2008) plus one day. Today's date was not used because the data ends in 2008 and every Recency value would otherwise be inflated by years.

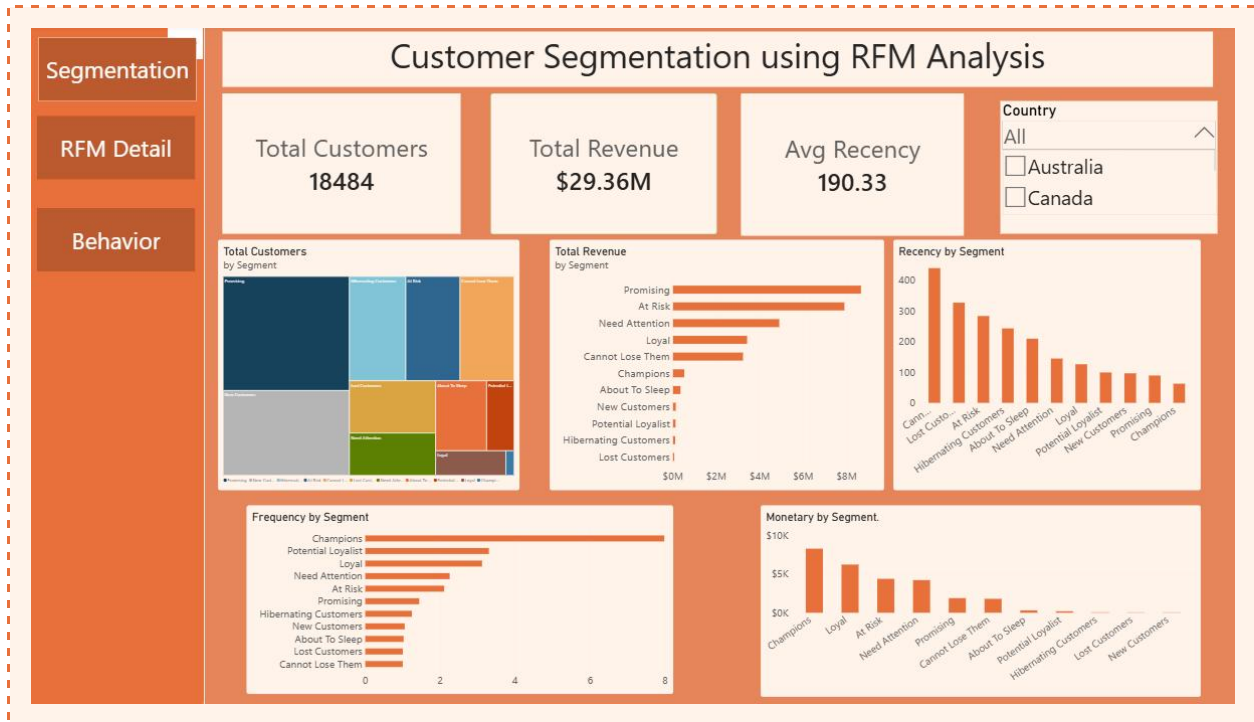
The RFM table was created in DAX and has one row per customer (18,484 rows). Recency ranges from 1 to 1,127 days. On average a customer has Recency 190 days, Frequency 1.5 orders and Monetary value of \$1,588.

6. Scoring and Segmentation

- **Recency and Monetary:** each was scored from 1 to 5 using the 20th, 40th, 60th and 80th percentiles. A more recent customer gets a higher Recency score, and a higher spender gets a higher Monetary score.
- **Frequency:** scored directly as 1 order = 1, 2 orders = 2, and so on, with 5 or more orders = 5. Percentiles were not used here because 62.9% of customers placed only one order, so the percentile bands would overlap.
- **RFM score:** the three scores were joined into one number (for example R=4, F=3, M=5 gives 435) and matched to a segment using the lookup table.

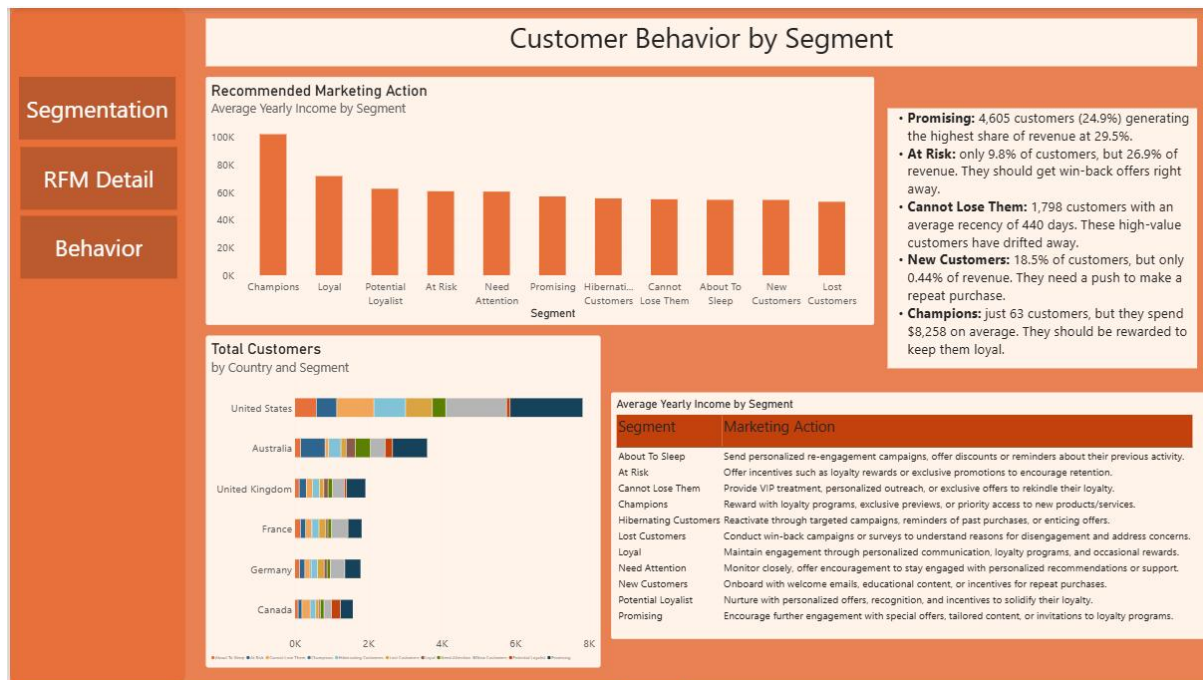
This produced 11 segments. Their size, revenue and average RFM values are shown below.

Segment	Customers	Customers %	Revenue %	Avg Recency (days)	Avg Freq.	Avg Spend
Promising	4,605	24.91%	29.50%	89.3	1.44	\$1,880
At Risk	1,807	9.78%	26.90%	283.9	2.11	\$4,370
Need Attention	1,165	6.30%	16.67%	144.8	2.25	\$4,201
Loyal	550	2.98%	11.63%	126.0	3.12	\$6,207
Cannot Lose Them	1,798	9.73%	10.98%	440.3	1.00	\$1,793
Champions	63	0.34%	1.77%	62.5	7.98	\$8,258
About To Sleep	1,142	6.18%	1.18%	209.6	1.02	\$302
New Customers	3,419	18.50%	0.44%	96.5	1.05	\$38
Potential Loyalist	611	3.31%	0.40%	99.4	3.30	\$191
Hibernating Customers	1,884	10.19%	0.34%	243.4	1.25	\$54
Lost Customers	1,440	7.79%	0.20%	327.4	1.00	\$41
Total	18,484	100%	100%	190.3	1.50	\$1,588



- **Promising and New Customers** are recent (about 90 to 97 days) but have bought only once or twice. Promising customers spend enough to be the largest revenue group; New Customers spend only \$38 on average.
- **Hibernating and Lost Customers** have low spend (\$54 and \$41) and long inactivity, so they contribute almost nothing to revenue.

In the demographic view of the dashboard, Champions and Loyal customers have the highest average yearly income. The United States has the largest customer base, followed by Australia, the United Kingdom, France, Germany and Canada.



8. Marketing Recommendations

Segment	Recommended action
Champions	Reward with loyalty perks, early access to new products and referral incentives. Keep them engaged, since they are few but spend the most.
Loyal	Upsell premium products, offer a loyalty programme and ask for reviews and referrals.
Potential Loyalist	They buy repeatedly but spend little. Use cross-sell offers and product bundles to raise the basket size.
Promising	Largest revenue group. Send follow-up offers and personalised recommendations to move them to a third and fourth purchase.
New Customers	Send a welcome email, onboarding content and a small incentive for a second purchase, because most never return.
Need Attention	Send personalised reminders and limited-time offers before they become inactive; they still bring 16.7% of revenue.
About To Sleep	Send reactivation reminders and discounts on categories they bought before.
At Risk	Top priority. Run a win-back campaign with personalised outreach, exclusive offers and a short feedback survey, since they hold 26.9% of revenue.
Cannot Lose Them	Give VIP treatment: direct personal outreach and special offers, because they were high spenders who have gone quiet.

Segment	Recommended action
Hibernating Customers	Use low-cost email reactivation only. Avoid heavy discounts because their average spend is small.
Lost Customers	Use a low-cost survey to learn why they left. Do not spend much on them; they bring 0.2% of revenue.

9. Key Insights

- Revenue is concentrated: five segments generate about 95.7% of sales.
- About 26% of customers (At Risk, Cannot Lose Them and Need Attention) hold about 55% of revenue and are drifting away or need attention. Retaining them is worth more than acquiring new customers.
- The largest customer group by count (Promising, then New Customers) has recent buyers with low frequency. Converting them into repeat buyers is the biggest growth lever.
- Only 0.3% of customers are Champions, which shows how few customers become highly loyal.

10. Limitations

- 62.9% of customers have only one order, so Frequency scores are mostly 1 or 2 and the top segments are small.
- The data is from 2005 to 2008, so the findings show the method rather than the current market.
- Recency and Monetary scores use percentile bands, so tied values can fall on either side of a band edge and results may differ slightly between tools.
- Amounts are in US dollars.

11. Conclusion

RFM analysis separated 18,484 customers into 11 clear segments and showed where revenue comes from and where it is at risk. The most useful actions are to win back At Risk and Cannot Lose Them customers, move Promising and New Customers to repeat purchases, and reward Champions and Loyal customers. The Power BI dashboard lets any of these groups be filtered by country and reviewed in detail.